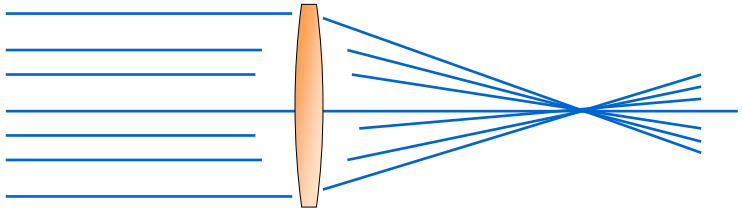
PHYC10009 Foundations of Physics

Imaging

15



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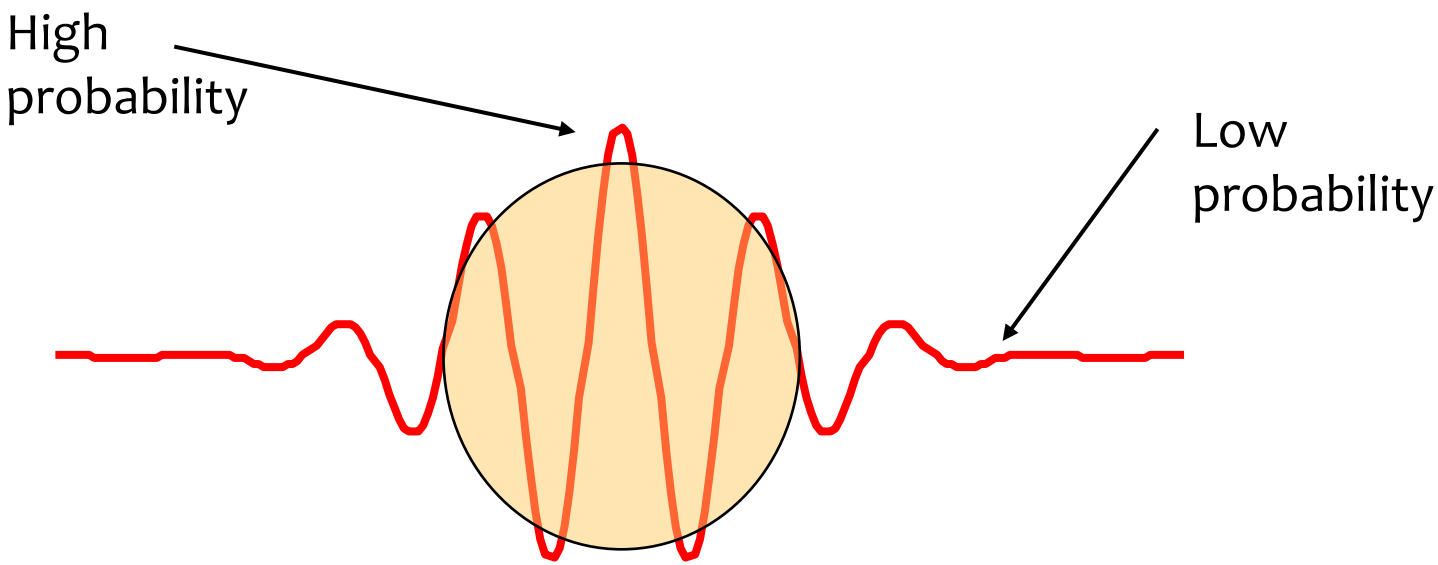
Quantum waves

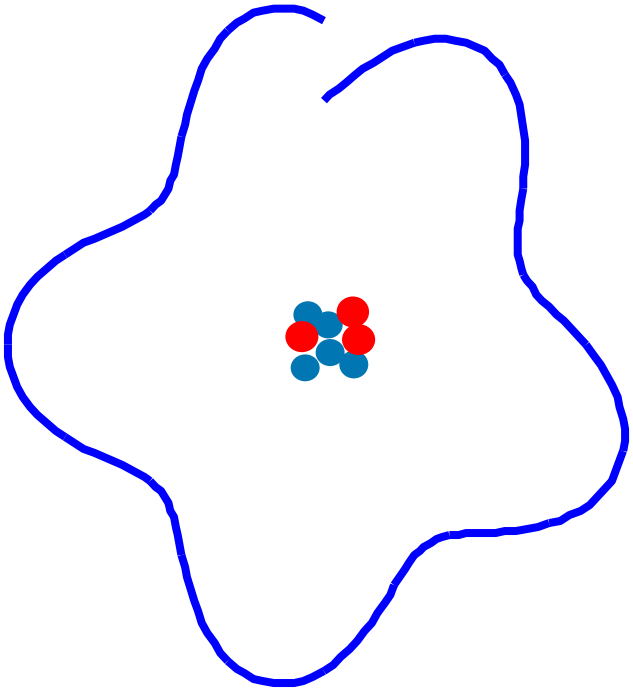
- Water wave = height of water
- Electromagnetic (light) wave = strength of field
- Quantum wave = *probability*
- Disturbance $D(x,t) \rightarrow \psi(x,t) \equiv P(x,t)$ is probability amplitude for finding the particle at position x at time t i.e. at (x,t)
- Wavelength of free particle $\lambda = \dfrac{h}{p}$ for momentum p



Electron waves and atoms

- Only certain momenta p allowed (N. Bohr)

$$\lambda = h/p$$

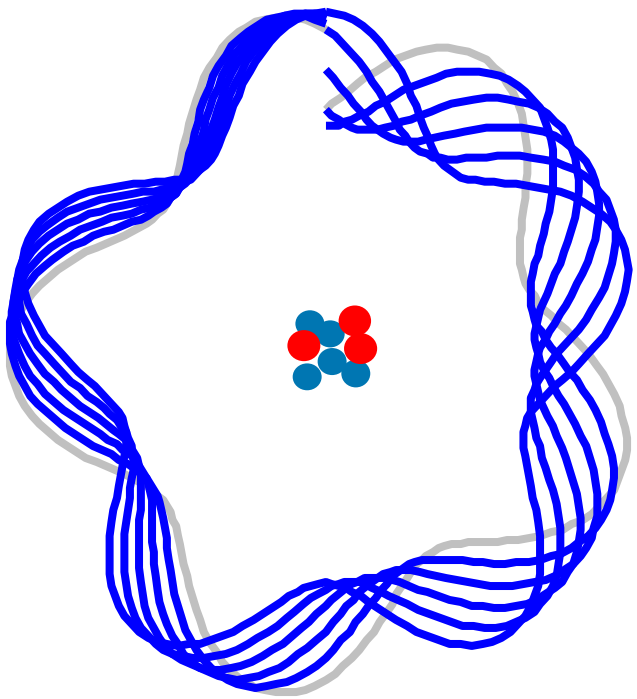


Electron waves and atoms

- Only certain momenta p allowed (N. Bohr)

$$\lambda = h/p$$

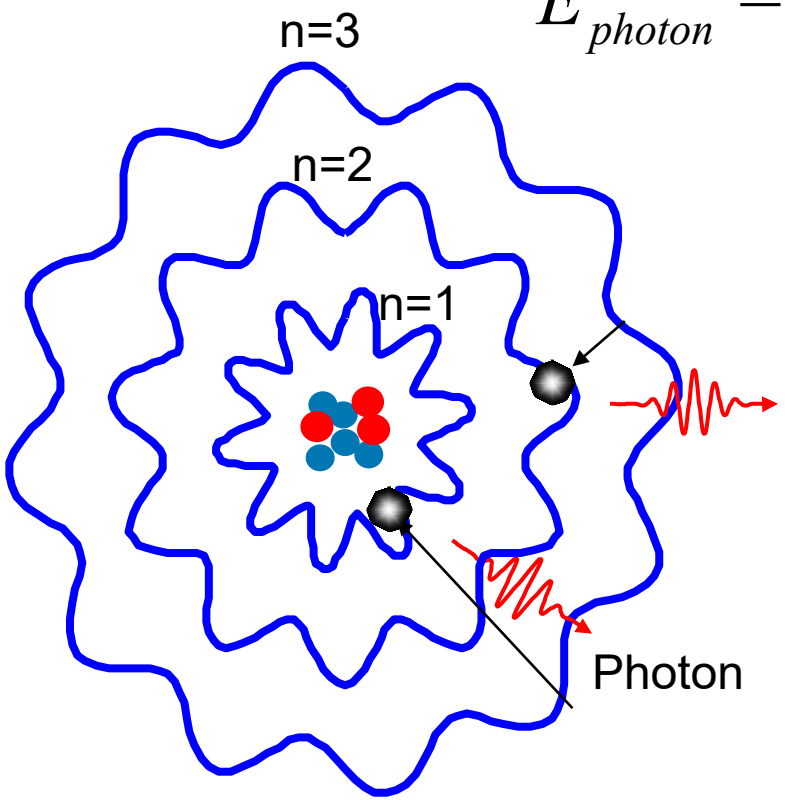
$p = 4.5$
$p = 4.6$
$p = 4.7$
$p = 4.8$
$p = 4.9$
$p = 5$



Quantum Atoms

- Discrete – *quantised* – levels explains atomic spectra and chemical shells

$$E_{photon} = hf = E_i - E_f$$



$$f = \Delta E / h$$

Summary

Waves

- Introduced "waves" as *propagating* disturbances $D\left(\frac{2\pi}{\lambda}x - \omega t\right)$
- Functions of <u>*two*</u> variables, e.g. x and t

$$D(x,t) = A\cos\left(\frac{2\pi}{\lambda}x - \omega t\right)$$

- Travelling waves; period, wavelength, *phase velocity* $v = \lambda f$

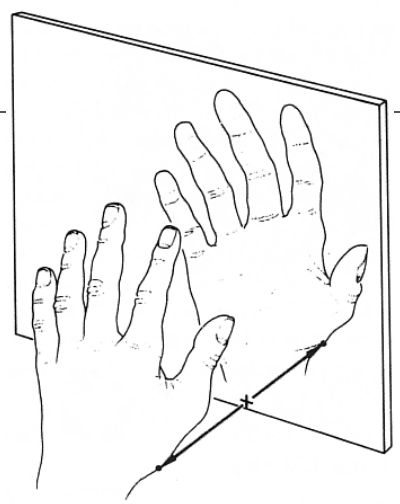
- Superposition

- Standing waves; resonance

Now

Light and optics

- 23.1 Light as rays
- 23.2 Light bounces: reflection
- 23.4 Light slows down: refractive index
- 23.5 Light bends: refraction; Snell's Law
- 23.6 Total internal reflection



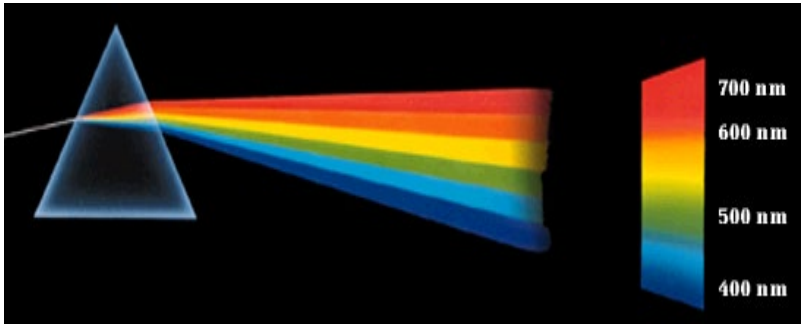
Chapter 23

Light and optics

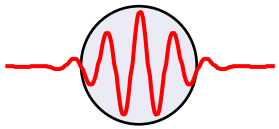
LIGHT AS RAYS

23.1 The ray model of light

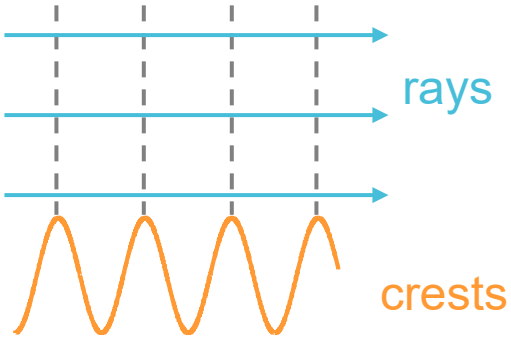
- For day-to-day optics, wavelength is ***short:*** $\lambda \sim 1\mu m$, smaller than objects of interest

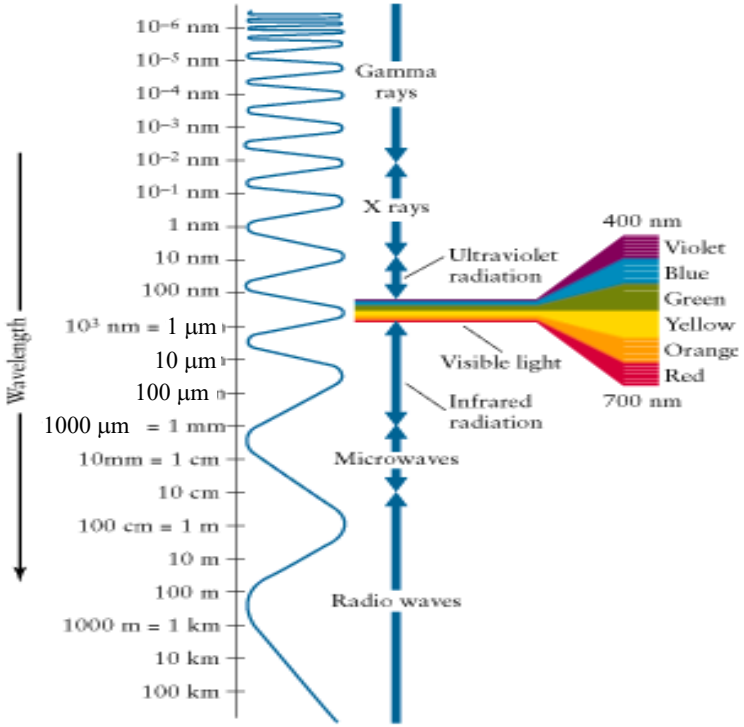


- Can treat waves as photons, or as ***rays***



- A *ray* is a line perpendicular to a wavefront
  - indicates direction of motion of wave
  - in this section we draw only rays

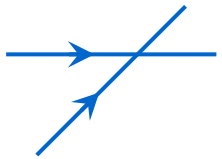


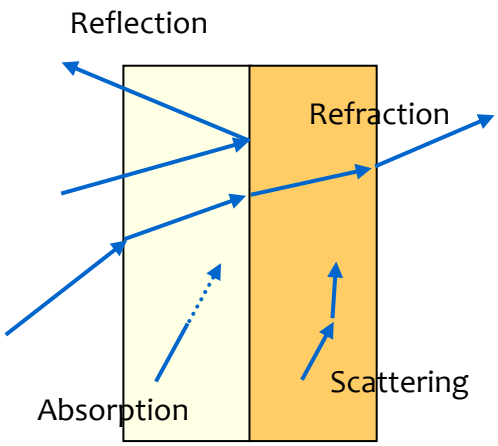


23.1 The ray model of light

The ray model

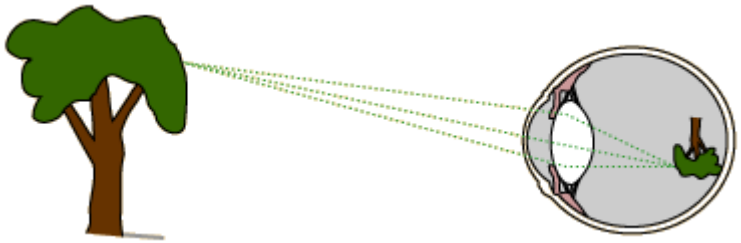
- Light rays travel in straight lines
- Light rays can cross
  (without interference)





- A ray travels forever until it interacts with matter

- An object (self-luminous <u>or</u> reflective) is a source of light rays
- The eye sees by focusing a diverging bundle of rays

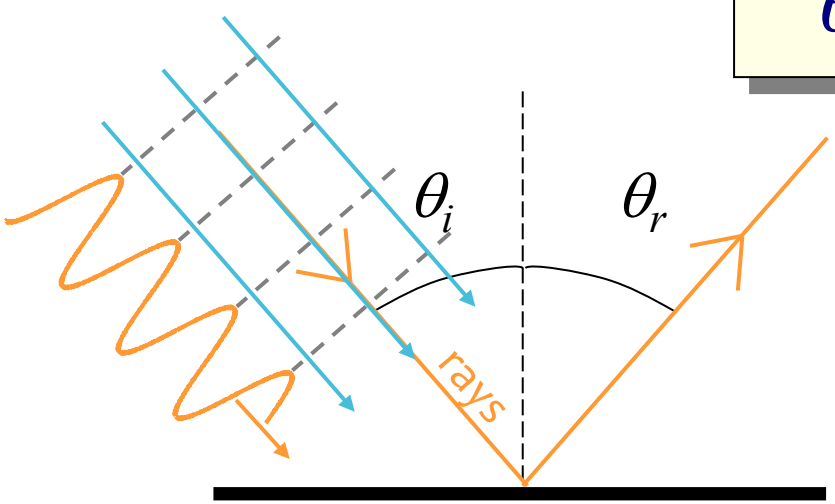


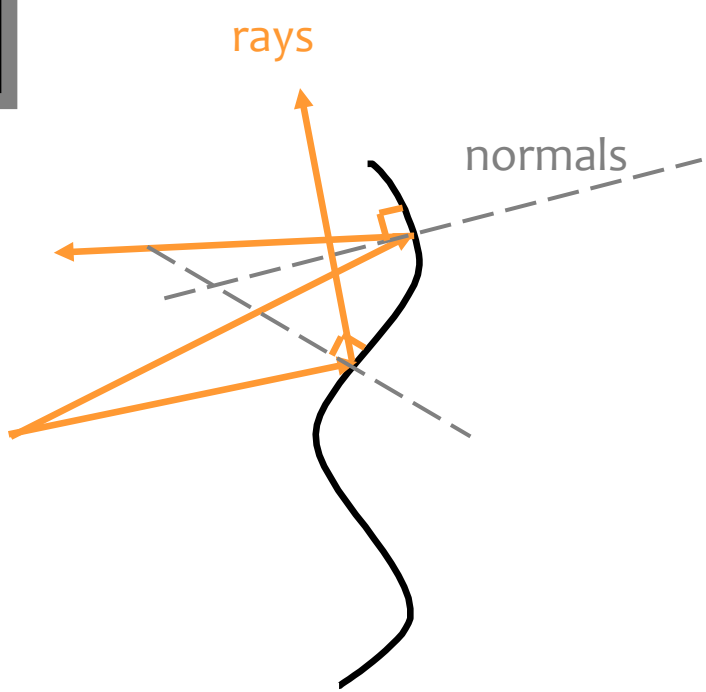
LIGHT BOUNCES: REFLECTION

23.2 Reflection

- Angle of reflection θ_r equals angle of incidence θ_i
- The incident ray, reflected ray and normal all lie in the *same plane*
- Note that angles are measured relative to ***surface normal***

$$\theta_i = \theta_r$$

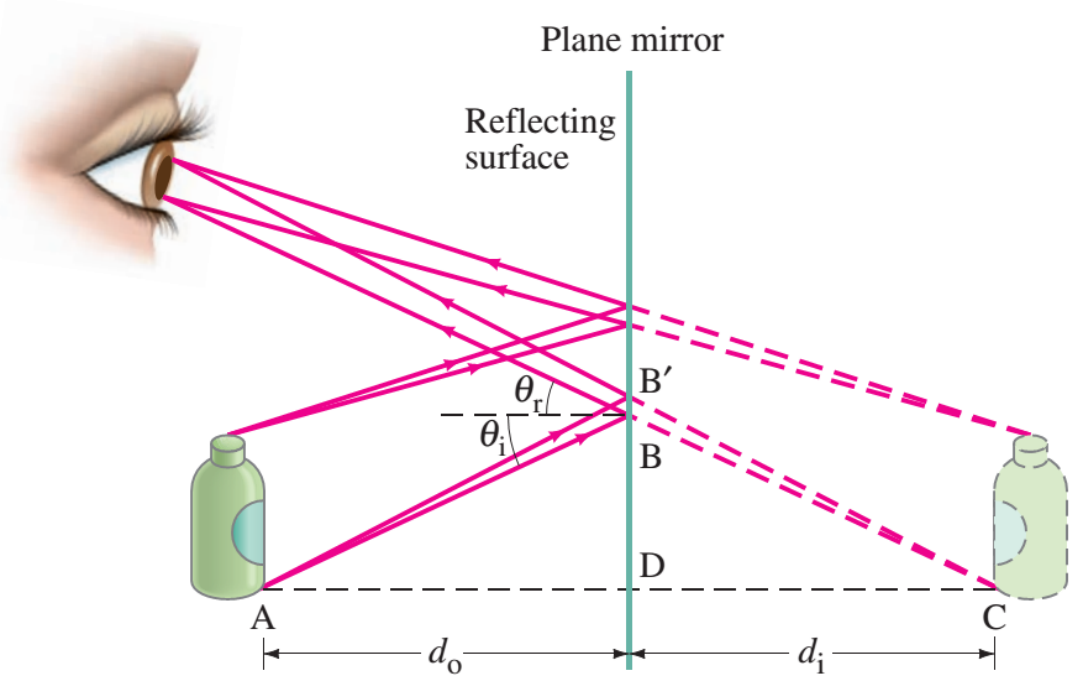


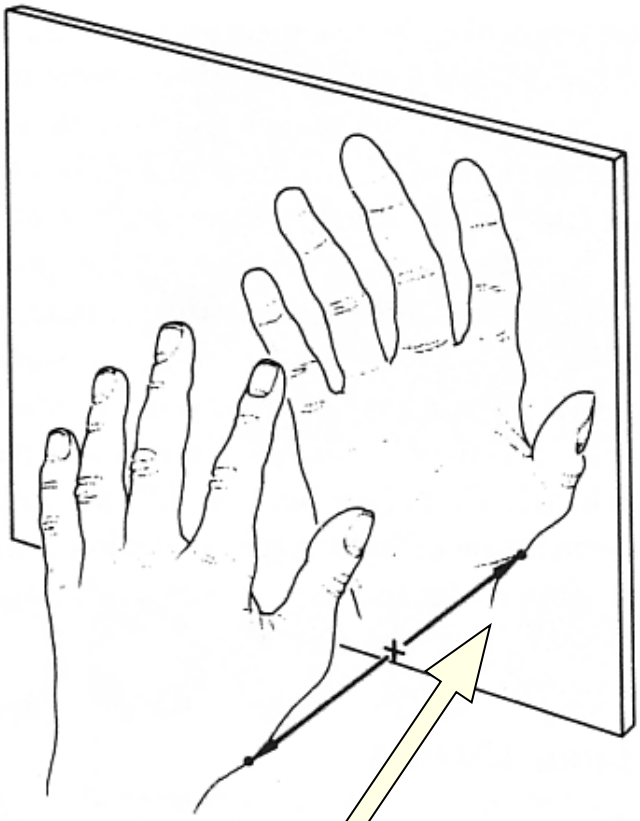


D Oa-6 Corner Reflector

Mirror inversion

- We see objects *behind* the mirror





Examples of *virtual* image

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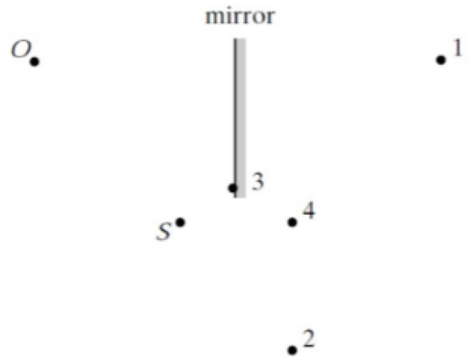
Location of Image

An observer *O*, facing a mirror, observes a light source *S*. Where does *O* perceive the mirror image of *S* to be located?



1

2

3

4

5 Some other location

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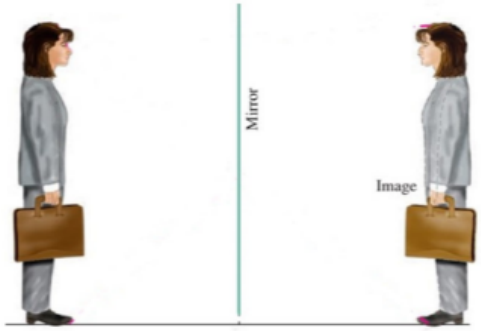
Size of mirror

A 2.0m tall person is standing 2.0 m from a flat vertical mirror staring at her image. What minimum height must the mirror's reflecting glass have if the person is to see her entire body, from the top of her head to her feet?



(a) 0.5 m

(b) 1.0 m

(c) 1.5 m

(d) 2.0 m

(e) 2.5 m

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Light and optics

- 23.1 Light as rays
- 23.2 Light bounces: reflection
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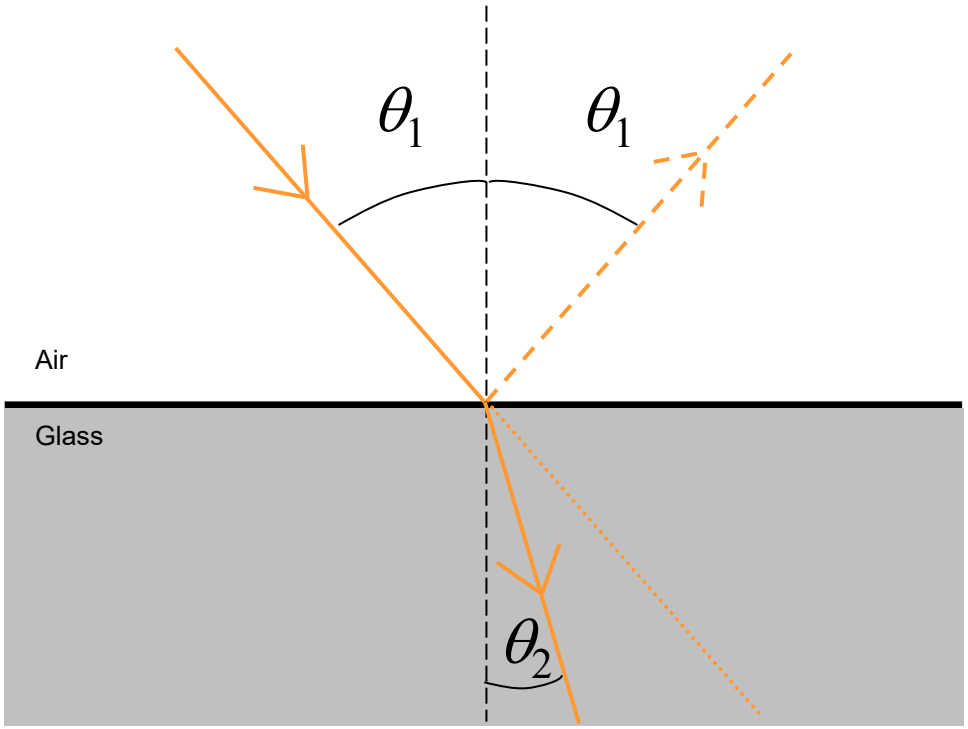
LIGHT BENDS: REFRACTION

23.5 Refraction

- Light propagating from one medium to another:
  - Partly reflected
  - Partly transmitted – *along a different direction*: REFRACTION
- Incident, reflected and transmitted rays, and the normal, all lie in the same plane
- But what is θ_2? Consider <u>speed</u> of light in different media.



LIGHT SLOWS DOWN: REFRACTIVE INDEX AND THE SPEED OF LIGHT

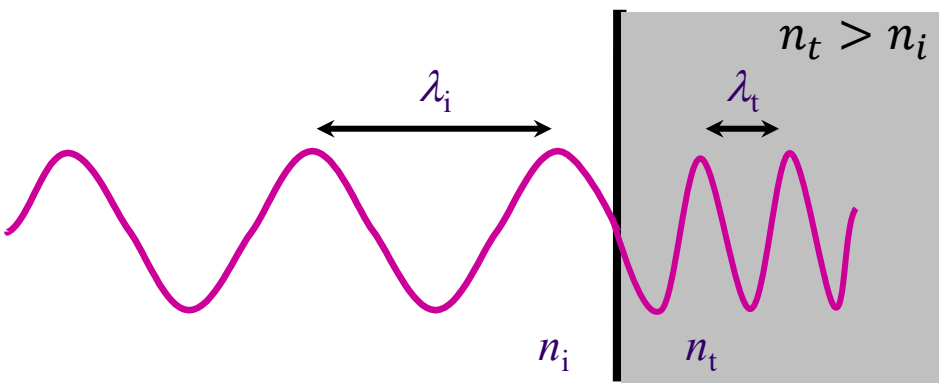
23.4 Speed of light: the index of refraction

- Velocity of light depends on material, just like sound

$$v = \frac{c}{n}$$ velocity of light in material

- n = ***index of refraction*** $(n > 1 \text{ so } v < c)$

- BUT: $v = f\lambda$ What happens to f, λ?
  - Frequency depends on source – does not change; wavelength must



$$v = \frac{c}{n} \qquad \lambda = \frac{v}{f} = \frac{c}{nf} = \frac{\lambda_0}{n}$$

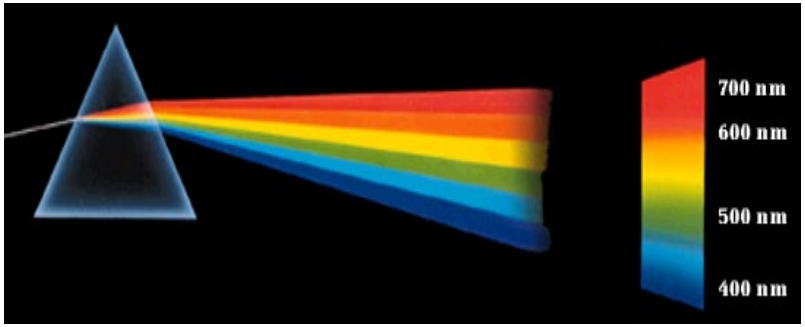
$$\frac{\lambda_i}{\lambda_t} = \frac{n_t}{n_i}$$

Remember: $\lambda = \frac{\lambda_0}{n}$

Lecture demo: matching refractive index

Refractive index

Material	Refractive index (λ=589nm)
Vacuum	1
Air (STP)	1.00029
Water	1.333
Glass	1.5 (varies)
Diamond	2.417
TiO_2	2.907
GaP	3.50
Atoms in special quantum state	Thousands
Bose-Einstein condensation	10,000,000 or more!

n is wavelength-dependent:

D Ob-1 Matching Refractive indices